D3R-Net: Denoising Diffusion-Based Defense Restore Network for Adversarial Defense in Remote Sensing Scene Classification

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***Abstract*— Deep learning models (algorithms) have demonstrated their superior performance in interpreting Earth science and remote sensing data. However, adversarial examples generated with perturbations imperceptible to humans could render deep learning algorithms ineffective. This significant vulnerability of deep learning models, thus, inspires the exploration of defense methods resistible to adversarial examples. Although numerous countermeasures against adversarial examples have been proposed, the design of a universally applicable defense method across multiple scenarios still remains to be explored. In this study, we propose an effective denoising diffusion-based defense restore network (D3R-Net) based on the denoising diffusion model from the perspective of adversarial restoration, which transforms the adversarial examples into clean samples. Utilizing a highly effective denoising diffusion probabilistic model (DDPM), our D3R-Net transforms input adversarial examples into a state of noise, where diverse forms of adversarial noise transition into Gaussian noise. Subsequently, it captures semantic information through a series of iterative denoising steps. The pixel distribution of adversarial examples is restored in the proposed network to match the original distribution, enabling the classifier to identify adversarial examples correctly. Furthermore, we introduce a combined filtering module to preserve the semantic information of the original image, thereby further enhancing the defensive performance. Instead of modifying the model structure or excluding suspected samples, the proposed method restores the adversarial examples, making it simple yet effective and applicable to a broader range of scenarios. Extensive experiments are conducted on four benchmark datasets, and the results demonstrate that D3R-Net has significant defense capabilities against known and unknown attacks. Our source code is available at https://github.com/SIM-xidian/D3R-Net.**

***Index Terms*— Adversarial defense, denoising diffusion, image restore, remote sensing image scene classification.**

I. Introduction

In RECENT decades, the swift progress in remote sensing satellite technology has supplied researchers with a rich array of diverse remote sensing imagery (RSI), encompassing varied spectral, temporal, and spatial resolutions. The abundant information in these RSIs has been utilized to develop more precise methods for Earth observation research. Among the fundamental tasks of interpreting RSI, scene classification plays an indispensable role. This task aims to accurately classify sub-regions extracted from the RSI into specific scene categories [1]. However, the high diversity within scene categories and the similarity between categories make this task challenging [2].

Early research on remote sensing scene classification tasks primarily focused on manually crafted features [3], [4] including shape, texture, color, spatial, and spectral features. However, the discriminative information in these features is limited, making it challenging to achieve high-precision classification. With the development of deep learning models (algorithms), such as VGG [5], Inception [6], ResNet [7], DenseNet [8], RegNetX [9], etc., their outstanding capabilities of feature extraction have been widely applied in remote sensing.

Deep convolutional neural networks (DCNNs) can learn representative and discriminative features from remote sensing images through convolutional layers. Compared to the low- and mid-level methods that relied on handcrafted features previously, DCNNs could not only overcome the difficulty of capturing complex relationships in the feature extraction process but also possess the advantage of end-to-end training models, enabling them to achieve higher accuracy in scene classification tasks [10]. The outstanding performance of DCNNs has led to the most advanced work in the field, such as the multimodal remote sensing image classification network and benchmark dataset [11], the remote sensing image super-resolution [12], and the spectral remote sensing foundation model SpectralGPT [13]. These latest works show that artificial intelligent techniques have become one of the universal technologies in remote sensing [14].

Although deep learning algorithms have achieved notable success in remote sensing classification tasks, they have also brought security concerns [15], [16], precisely their vulnerability to adversarial examples [17], [18], [19]. Adversarial examples refer to deceptive samples generated by applying slight perturbations to clean samples. As shown in Fig. 1, for the input clean remote sensing samples with correct labels, the trained model can generally classify them correctly. In contrast, adversarial examples are misclassified. These

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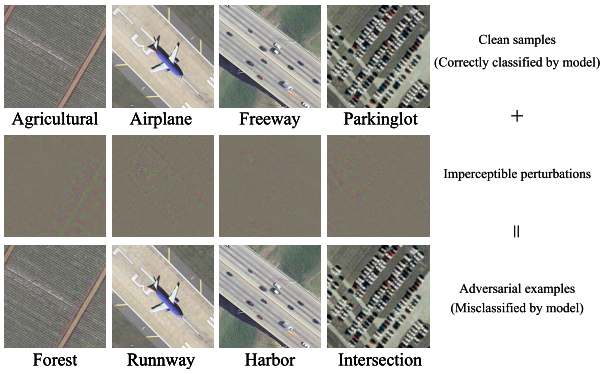


Fig. 1. Generation process of adversarial examples. Correctly classified clean samples are listed in the first row, and their corresponding adversarial examples, which are misclassified, lie in the last row. The slight differences between them are difficult to observe by humans but can be sensed by DCNN.

subtle differences between adversarial and original images may be imperceptible to human observers but can lead to the failure of trained models [20]. Given the high-security considerations in most Earth science and remote sensing tasks, it is paramount to develop adversarial defense methods and enhance the resistibility of deployed deep learning models (algorithms) against adversarial examples. These adversarial defense approaches aim to strengthen the robustness of models to enhance the resilience of adversarial examples [15], [21].

So far, several defense methods against adversarial examples have been proposed, which can be roughly divided into adversarial training [1], [22], [23], adversarial examples detection [20], [24], and adversarial examples restoration [21], [25], [26], all to mitigate the risk of adversarial examples' compromising deep learning models.

Among them, first, Cheng et al. [23] introduced an adversarial training strategy, which has been proven highly effective. However, the architecture of target model is required to be modified in this method, making it necessary to retrain the deployed model. In practical applications, since retraining the deployed model may not be feasible, it is crucial to develop an adversarial defense method, which could directly mitigate the impact of adversarial examples. Second, Chen et al. [20] conducted in-depth research on the differences between adversarial examples and clean samples, and then developed an adversarial examples detection strategy, which could detect and remove adversarial examples from the input data without retraining. Although the retraining issue is solved in this approach, the problem of reducing samples would emerge and may restrict its application in limited sample scenarios. Third, in addition, Xu et al. [21] developed a task-guided denoising defense network to transform adversarial examples into clean samples in a supervised manner, but its defensive performance is still insufficient.

Although these methods have some defensive effects, they could only be applied in specific scenarios. These methods have limitations, mainly when computational power is limited or data is sparse. For example, adversarial retraining and redeployment of models with limited computational resources are challenging. Adversarial retraining [23] relies on a large amount of labeled data for improvement, which is costly and time-consuming in practice. In addition, frequent model tuning reduces operational efficiency. Second, although the adversarial sample detection [20] method can detect many suspicious samples, these samples cannot be reused, which makes it challenging to apply when sample resources are scarce. Lastly, due to its adherence to the supervised learning framework and high dependence on large-scale labeled data, the current adversarial examples restoration [21] method based on deep learning faces the dual challenges of data acquisition bottlenecks and poor adaptability to source model attacks. It results in poor performance in actual adversarial defense. Therefore, seeking a more comprehensive and practical solution in adversarial environments to protect existing models from the influence of adversarial examples remains an unresolved problem.

Adversarial examples restoration or purification methodologies based on the denoising diffusion probabilistic model (DDPM) [27], [28] have emerged as a distinctive strategy in adversarial defense research, excelling in scenarios with limited computing resources and sample data. By adopting a principle of adversarial noise purification, DDPM circumvents costly retraining and complex sample elimination, instead employing a reverse denoising process to rehabilitate adversarial inputs and robustly counteract unseen attacks. The DDPM's efficacy is rooted in its dual strengths: generative prowess and a stepwise denoising mechanism. Through iterative forward diffusion, introducing Gaussian noise followed by precise reconstruction via neural networks in reverse, adversarial perturbations are effectively neutralized akin to noise removal. This cyclic denoising fosters a progressive restoration of corrupted data to its pristine form, thereby converting adversarial instances into benign ones.

By statistical analysis, the pixel distributions of adversarial examples are observed to be shifted compared to clean samples, which are caused by the introduced noise and result in misclassification. To tackle this issue, an adversarial defense method is designed in this article from the perspective of noise reduction restore for remote sensing scene classification, based on a proposed DDPM [27], [28]. In this method, in order to be classified correctly, the input adversarial examples are first diffused into the noise state for a certain number of steps, stepwise blend adversarial noise into noise under Gaussian distribution, and then restored through denoising process. Additionally, due to the complexity of remote sensing images with intricate backgrounds and rich semantic information, preserving semantic information is crucial for the classifier's accurate discrimination. Therefore, we have designed a *dynamic composite filtering module* (DCFM) to retain the semantic information of the clean samples. Our method constitutes an unsupervised vanguard classifier defense network that is trainable without reliance on sample labels. At its core lies the efficacious denoising capability of DDPM, enabling operation both in offline and real-time scenarios. The network effectively mitigates adversarial noise, rectifies the presence of adversarial samples within datasets, thereby bolstering the resilience of the classifier in accurately recognizing such instances, manifesting considerable potential for practical application. Through extensive experiments on multiple datasets, we validate the effectiveness of the proposed defense

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network against unknown attacks. The contributions outlined in this article can be succinctly summarized as follows.

1) A novel defense framework is proposed based on the DDPM. The model stands out with its powerful and stable generative capabilities and does not rely on training labels, which grants this method higher practical value.
2) An innovative design of the DCFM is introduced to preserve the rich semantic information of clean samples, which makes recovered images more refined and significantly enhances the defense capability of the method.
3) We devised two different applications of the proposed framework. Notably, the proposed network has almost no adverse effects on clean samples. As a result, it can be applied offline and seamlessly into online modes for real-time recognition by incorporating it into the classifier model.

The subsequent sections of this article are structured as follows. Section II briefly reviews the work related to this study. Section III elaborates on our defense approach in detail. Section IV presents the comprehensive results of the four benchmarks and conducts the analysis. Finally, SectionV provides a summary of this article.

II. Related Work

A. Adversarial Attack

1) Fast Gradient Sign Method: Fast gradient sign method (FGSM) is a white-box method of single-step attack, proposed by Goodfellow et al. [29]. It uses the gradient information of the neural network to generate adversarial examples, causing clean samples to be slightly perturbed, thereby leading to misclassification. Specifically, adversarial example x_{adv} is made by perturbing each pixel of the input image x in the gradient direction. The adversarial example x_{adv} can be expressed as follows:

$$x_{\text{adv}} = \text{clip}(x + \epsilon \text{sign}(\nabla_x J(\theta, x, y))) \tag{1}$$

where $\nabla_x J(\theta, x, y)$ represents the gradient of the loss function $J_{(\cdot)}$ computed for the input x, and y represents the true label of the sample input x. sign$(\cdot)$ represents a sign function. clip$(\cdot)$ represents the pixel value in the clipped image, and ε is a small scalar value controlling the norm of the perturbation.

2) Project Gradient Descent: For gradient-based attacks, project gradient descent (PGD) [30] can be viewed as an iterative version of FGSM. It generates adversarial examples by iteratively applying gradient descent. In each iteration, the gradient of the loss function corresponding to the original input is calculated with respect to the input, and then the input is slightly adjusted in the opposite direction of the gradient. This process is repeated multiple times until the generated adversarial examples are able to fool the model into misclassification. The adversarial example x_{t+1} is calculated by PGD as

$$x_{t+1} = \prod_{x+s}(x_t + \alpha \cdot \text{sign}(\nabla_x J(\theta, x, y))). \tag{2}$$

By setting the parameters α and s of attack iterations through PGD, adversarial examples with different attack strengths can be obtained.

3) Carlini and Wagner: The Carlini and Wagner (C&W) adversarial attack method was proposed by Carlini and Wagner [31]. Its main idea is to minimize the difference between adversarial examples and original input while maintaining the misclassification of adversarial examples. The C&W adversarial attack method can be divided into two parts: generating adversarial examples and minimizing the adversarial loss function. Generating adversarial examples is achieved by adding small perturbations to the original input, so the deep neural network produces incorrect classification results. Minimizing the adversarial loss function is to reduce the difference between adversarial examples and original input, which enables the adversarial examples to fool the deep neural network. C&W can be expressed as

$$\arg\min_{x_{\text{adv}}} \|x_{\text{adv}} - x\|_\infty - \mu J(\theta, x_{\text{adv}}, y) \tag{3}$$

where μ is the weighting factor.

4) Mixup-Attack and Mixcut-Attack: Xu and Ghamisi [32] discovered common vulnerabilities between different networks by attacking the shallow features of a given agent model and developed universal adversarial examples (UAE). This method uses the mixup and cutmix strategies to generate attack sample images, which are mixed with clean samples during the attack to be classified incorrectly.

B. Adversarial Defense

1) Adversarial Training: Adversarial training [1], [22], [23] uses both adversarial examples and clean samples to retrain the classifier. It is one of the earliest methods to defend against adversarial examples and has been widely used in many previous studies. Along with these two types of samples, Cheng et al. [23] also introduced the samples generated during image reconstruction to train the classifiers and developed a new adversarial training framework.

2) Adversarial Examples Detection: Adversarial examples detection [20], [24] aims to learn the difference between adversarial samples and clean samples, and uses a certain strategy to mark suspicious samples as adversarial samples for elimination. Chen et al. [20] proposed a soft threshold defense method, which takes the output confidence of the sample as input and obtains the decision boundary through a logistic regression algorithm. In adversarial defense, samples whose confidence thresholds output by the deep learning model are higher than the corresponding category thresholds are classified as clean samples; otherwise, they are adversarial examples.

3) Adversarial Examples Restoration: It regards adversarial attacks as the destruction of samples. In defense, it is easy to think of restoring the damaged samples [21], [25], [26]. Meanwhile, the most related work to our method is a novel task-oriented denoising network (TGDN) proposed by Xu et al. [21] for adversarial defense for remote sensing scene classification. In this method, a denoising network is trained using clean remote sensing samples and the corresponding adversarial examples. By denoising the samples before they are fed to the deep learning model, the network significantly could improve its resistance to adversarial examples.

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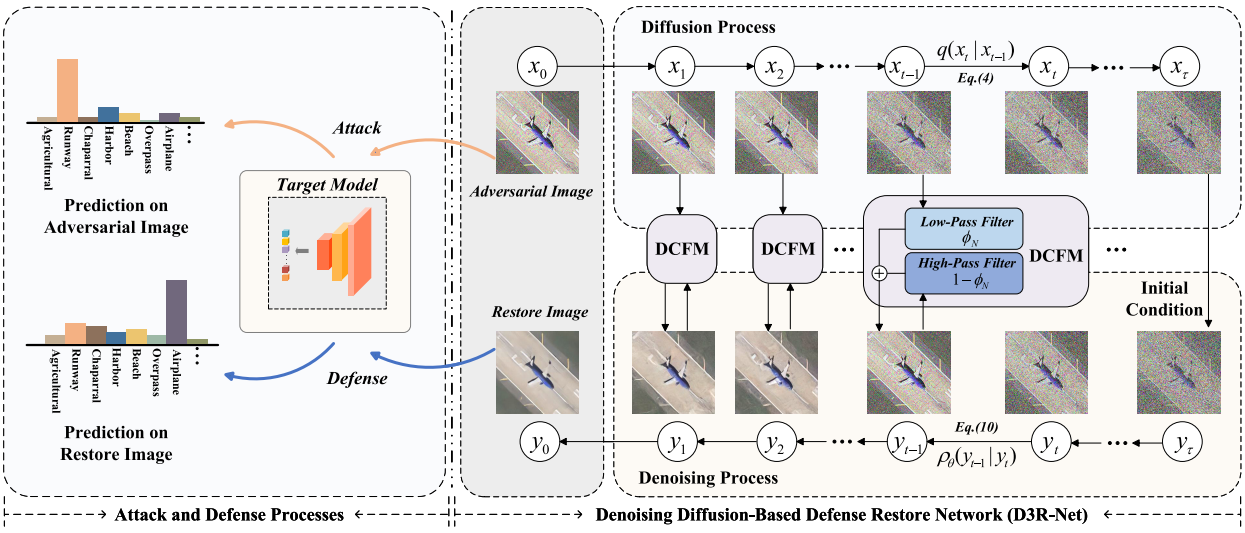


Fig. 2. Overall framework of the proposed method. The left part shows the attack and defense processes. Specifically, it is the process of attacking the target model with adversarial samples to make the target model misclassify and the process of reconstructing the sample defense so that the target model is classified correctly. The right part shows our defense framework for converting an adversarial image x_0 into a restored image y_0. The D3R-Net consists of the forward diffusion and reverse denoising processes. In the reverse process, we introduce the DCFM.

C. Diffusion Models

Diffusion probabilistic models are a rapidly evolving class of generative models that compete with generative adversarial networks (GANs) [33] in unconditional image generation. The diffusion model comprises of two interrelated processes: a predefined forward process and a corresponding reverse process. The forward process transforms the data distribution into a simpler prior distribution, such as a Gaussian distribution. In contrast, the reverse process gradually undoes the forward process by simulating ordinary or stochastic differential equations (ODE/SDE) [34]. The forward process resembles a simple Brownian motion with time-varying coefficients. Afterward, the neural network is trained to estimate the score function using the denoising score-matching objective. Compared to adversarial targets in GANs, diffusion models provide more stable training targets, and compared to methods, such as variational autoencoders (VAEs) [35], energy-based models (EBMs) [36], and normalized flows, diffusion models exhibit better generation quality. Recent studies have shown that some diffusion models have been improved and widely used in various tasks, such as ILVR [37] and TDPM [38]. Remote sensing images are large and have complex backgrounds. Therefore, inspired by the powerful generative ability of diffusion models, we apply them to the adversarial defense task of remote sensing image classification. We improve and optimize it based on the same hybrid architecture as TDPM.

III. Proposed Method

In this section, the details of our proposed defense framework are introduced, including forward diffusion process, reverse denoising process, training objective, and the adaptive composite filtering module. The overall architecture of the proposed defense framework is shown in Fig. 2, which mainly consists of two parts, namely attack and defense processes and denoising diffusion-based defense restore network (D3R-Net).

A. Overview

A noisy image is a condition unrelated to degradation during DDPM generation. Compared with the pixel distribution of clean samples and adversarial examples before diffusion, it is more indistinguishable after adding a certain number of steps of Gaussian noise. However, DDPM can still capture the semantic information in this noisy state and restore it to be consistent with the original distribution, that is, restore the adversarial examples aligned to clean samples. Fig. 2 illustrates the overall framework of the proposed defense framework. x_0 denotes the adversarial example image; meanwhile, y_0 denotes the restore image corresponding to x_0. The process of adversarial attack and adversarial defense is shown on the left. The adversarial example restoration process, specifically D3R-Net, is shown in the right part of Fig. 2. Two processes are required to obtain the restored adversarial examples, namely, the diffusion process and the denoising process. The left side of Fig. 2 simulates the two processes of attack and defense. In the attack context, the adversarial example x_0 is directly presented to the target model for recognition, leading to a misclassification, constituting the attack process. On the right side of Fig. 2, x_0 is processed through the diffusion and denoising processes, ultimately restored as y_0. Introducing y_0 into the target model results in an accurate solution to the classification problem, which marks a successful implementation of the defense process.

By adding Gaussian noise to adversarial examples, the diffusion process makes the pixels distribution of clean and adversarial examples indistinguishable from each other, that is, the adversarial noise is fused into Gaussian noise. Therefore, a denoising process is integrated subsequently to remove adversarial noise from the images at each step.

In the denoising process, to better preserve the semantic information of images, DCFM is proposed and used in each step of the denoising process. The DCFM aims to eliminate

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noise while dynamically adjusting filter parameters according to the image noise proportion so that the semantic information of the reconstructed images is maintained as much as possible.

Combining these two processes enables D3R-Net to effectively restore adversarial examples to a state consistent with the original distribution in the presence of Gaussian noise. More details are discussed below.

B. Denoising Diffusion-Based Defense Restore Network

1) Forward Diffusion Process: DDPM constitutes a category of generative models. A variance table is first predefined as $\{\beta_1, \beta_2, \ldots, \beta_T\}$, and images are then gradually corrupted into a noisy state with a forward (diffusion) process

$$q(x_t|x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t x_{t-1}}, \beta_t \boldsymbol{I}\right). \tag{4}$$

Furthermore, based on the properties of Markov chains, for any intermediate time step $t \in \{1, 2, \ldots, T\}$, the corresponding noise distribution has an analytical form

$$\begin{aligned} q(x_t|x_0) &= \mathcal{N}\left(x_t; \sqrt{\overline{\alpha}_t}x_0, (1-\overline{\alpha}_t)\boldsymbol{I}\right) \\ &= \sqrt{\overline{\alpha}_t}x_0 + \sqrt{1-\overline{\alpha}_t}\epsilon \end{aligned} \tag{5}$$

where $\overline{\alpha}_t := \prod_{s=1}^{t}(1-\beta_s)$ and $\epsilon \sim \mathcal{N}(0, I)$. If T is large enough, $x_T \sim \mathcal{N}(0, I)$, usually $T = 1000$.

For the proposed framework, we first model the restore process of adversarial examples as $p_\phi(y \mid x)$. This process aims to restore the clean sample y_0 from the adversarial examples x_0, which can be formulated by maximizing the likelihood

$$p_\phi(y \mid x) = \int p_\phi(y_0 \mid x_0) dx_0. \tag{6}$$

In the diffusion process defined in (4) for a sufficiently large t, $t \approx T = 1000$, even two completely different images x and y could be exactly the same after t-step diffusion, that is, both $q(y_t \mid y)$ and $q(x_t \mid x)$ are naturally close to the standard Gaussian distribution. Given that there is a precondition for adversarial examples, that is, the adversarial examples are visually indistinguishable from the clean samples. According to the assumption in [39], there is an intermediate time step τ, so that for $\tau < T$, the distance between $q(y_\tau \mid y)$ and $q(x_\tau \mid x)$ is finally small enough to satisfy $q(y_\tau \mid y) \approx q(x_\tau \mid x)$. Based on this, we rewrite the objective of the adversarial example restore procedure by applying the assumption $q(y_\tau \mid y) \approx q(x_\tau \mid x)$

$$\begin{aligned} p_\phi(y \mid x) &= \int p_\theta(y_0 \mid x_\tau) q(x_\tau \mid x_0) dx_0 dx_\tau \\ &\approx \int p_\theta(y_0 \mid y_\tau) q(y_\tau \mid x_0) dx_0 dy_\tau \end{aligned} \tag{7}$$

$$p_\theta(y_0 \mid y_\tau) = \prod_{t=1}^{\tau} p_\theta(y_{t-1} \mid y_t). \tag{8}$$

By substituting the variable from x_τ to y_τ. Equations (7) and (8) naturally lead to a DDPM model that denoises y_τ back to y_0. Specifically, we first "forward" the adversarial example x_0 to the initial condition x_τ by sampling from (4), and use it as the initial condition y_τ for the reverse denoising process

$$y_\tau := x_\tau = \sqrt{\overline{\alpha}_\tau}x + \sqrt{1-\overline{\alpha}_\tau}\epsilon \tag{9}$$

where $\tau \in \{1, 2, \ldots, T\}$. This corresponds to $q(y_\tau|x_0)$ in (7). We set T to 100 in the experiments. Then the denoising process starts from step τ. This reduces sampling steps and also helps speed things up.

2) Reverse Denoising Process: Given the above forward process, the model gradually generates images by reversing the forward process. The generative process is also a Gaussian transition with the learned mean μ_θ

$$p_\theta(y_{t-1}|y_t) = \mathcal{N}\left(y_{t-1}; \mu_\theta(y_t, t), \sigma_t^2 I\right) \tag{10}$$

where σ_t is usually a predefined constant associated with the variance table and $\mu_\theta(y_t, t)$ is usually parameterized by the denoising U-Net [40]. $\mu_\theta(y_t, t)$ with the following equivalence:

$$\mu_\theta(y_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(y_t - \frac{1-\alpha_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon_\theta(y_t, t)\right). \tag{11}$$

3) Training Objective: Similar to DDPM, we parameterize a denoising U-Net [40] model as $\mu_\theta(y_t, t)$. U-Net has a simple structure and can effectively capture contextual information to improve the quality and generalization ability of generated images. L_{t-1} can be viewed as a loss between $\ell 2$-two averaged coefficients

$$L_{t-1} = \mathbb{E}_q\left[\frac{1}{2\sigma_t^2}\|\tilde{\mu}_t(y_t, y_0) - \mu_\theta(y_t, t)\|^2\right] + C. \tag{12}$$

Among them, C is the constant term and $\tilde{\mu}_t(y_t, y_0)$ means that y_0 is the output of step t under the initial conditions.

C. Dynamic Composite Filtering Module

In order to retain the rich semantic information of remote sensing images and enhance the discriminability features, we designed a DCFM. Specifically, during the reverse denoising process, we replace the low-frequency part of y_{t-1} with the bottom-frequency part of x_{t-1}, and the formula is as follows:

$$x_{t-1} := \Phi_N(y_{t-1}) + (\mathbf{I} - \Phi_N)(x_{t-1}) \tag{13}$$

where $\Phi_N(\cdot)$ denotes a low-pass filter implemented by downsampling and upsampling the samples. The basic idea of this method is to first remove high-frequency information through downsampling, and then reconstruct the signal through upsampling and interpolation, thereby achieving the effect of low-pass filtering, where N is the scale factor for downsampling. During diffusion, the noise proportion increases as the depth increases, and the available information decreases gradually, so a more prominent sampling factor N is required.

In the design of the factor N, we account for the varying noise proportion throughout the diffusion process: increasing during diffusion and decreasing during denoising. At high noise levels, such as step $\tau = 100$, where high frequencies primarily represent noise, a larger N is mandated to preserve low-frequency components and extract semantic information. Conversely, lower noise levels necessitate a smaller N. Thus,

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we dynamically link N to the noise proportion of the image, using the original image x_0 as a baseline to estimate noise content and guide the selection of N—large N for high noise, small N for low, adjusting N dynamically at every step to mirror the changing noise profile of the image. For N, we design as follows:

$$N = 2^{\lfloor \mathrm{RMSE}(x_{t-1}, x_0) \rfloor}. \tag{14}$$

We measure the noise proportion of the image by calculating the root mean square error (RMSE) between x_{t-1} and x_0. At this time, N is controlled by the noise proportion. When the noise is large, a larger N is used, otherwise a smaller N is used, which is dynamically adjusted. The DCFM is crucial to preserve the low-frequency information of the input samples. The distribution consistency between adversarial examples and clean samples could be better preserved through iterations, thereby improving the quality and accuracy of image restoration.

IV. Experiments

A. Dataset

In this article, three benchmarks and one publicly available remote sensing adversarial example benchmark are used to verify the proposed framework, including the UC Merced dataset [41], the AID dataset [42], the NWPU RESISC45 dataset [43], and the UA-RSE dataset [32].

1) UC Merced Dataset: Images in the UC Merced dataset [41] are from the United States Geological Survey (USGS) national map. There are a total of 21 land use categories in the dataset, each of which contains 100 images in red, green, and blue (RGB) color space. These images are 256 × 256 pixels in size and have a resolution of 0.3 m.

2) AID Dataset: The images in the AID [42] dataset come from Google Earth. AID is a large-scale dataset for RSI scene classification. It contains a total of 30 categories, each category contains 220–420 images, and a total of 10 000 images. The images are 600 × 600 pixels in size and have resolutions ranging from 8 to 0.5 m.

3) NWPU-RESISC45 Dataset: The images in the NWPU-RESISC45 dataset [43] are from Google Earth. It contains 45 categories with 700 images in each category, for a total of 31 500 images. All images were 256 × 256 pixels in size, with resolutions ranging from 30 to 0.2 m per pixel. The number of categories and the total number of images in this dataset are very large. In addition, in the RSI scene classification task, this dataset is more challenging due to its intra-class diversity and inter-class similarity characteristics.

4) UA-RSE Dataset: UA-RSE dataset [32] is the first publicly available remote sensing adversarial examples dataset, which has two distinct types of adversarial examples datasets: UAE-RS UCM and UAE-RS AID. For these three datasets, UC Merced, AID, and NWPU-RESISC45, we randomly select a fixed 50% of samples for training the target model. This part of the samples is invisible to both our proposed method and the comparison method and is only used to train the target model. The remaining 50% of samples are used as known samples for training and testing. The images were resized to 256 × 256 pixels in experiments. All experiments were implemented on the PyTorch platform using NVIDIA RTX 3080TI (12G) GPU.

TABLE I
Classification Accuracy (%) of All Models on the Three Datasets

Model	UCM	AID	NWPU
VGG16	91.90	89.62	86.05
Inception-v3	95.33	93.70	91.91
ResNet18	95.90	93.74	90.96
ResNet50	94.09	92.46	91.47
ResNet101	95.42	91.58	91.18
DenseNet121	96.57	95.04	92.97
DenseNet201	96.19	94.00	92.85
RegNetX-400MF	96.76	94.40	92.80

B. Experimental Settings

For the evaluation metrics, to make the results more fair and intuitive, classification accuracy is used as the evaluation index of defense performance (defense accuracy), that is: $\mathrm{DA} = n_{\mathrm{correct}}/n_{\mathrm{total}}$ is used as the evaluation index, where n_{correct} represents the number of correctly classified samples of adversarial examples, and n_{total} is the total number of samples. A higher DA indicates a more robust defense.

In this section, we use VGG16, Inception-v3, ResNet18, ResNet50, ResNet101, DenseNet121, DenseNet201, and RegNetX-400MF as the target model, which are commonly used and show excellent performance, and their classification results on clean samples are shown in Table I.

A thorough experiment can be broken down into four main parts. The first part is the defense experiment of different attack methods, and some commonly used methods are used, such as FGSM [29], IFGSM [44], PGD [30], C&W [31], Jitter [45], Mixup-Attack [32], and Mixcut-Attack [32]. We use ResNet18 as the source model to generate the corresponding adversarial examples and fix the disturbance level and step size in all methods to 1. We fixed the total number of iterations for the iterative method to 5. In order to verify the defense capability against destructive attacks, the defense performance against different destructive attacks is given in the ablation research section. All methods in this section use the ∞ norm to compute against perturbations. We first evaluate the defense capability of the proposed defense method on attacks.

The second part presents the comparative experiments conducted on the first publicly available remote sensing adversarial examples dataset [32], UA-RSE. UA-RSE has two distinct types of adversarial examples datasets: UAE-RS UCM, and UAE-RS AID. We use nine comparative methods: two traditional filtering defense methods, Mean-Filter [46] and GB-Filter [47]; two generative defense methods, APE-GAN [22] and MAD-VAE [48]; two deep learning-based denoising defense methods, HiNet [49] and TGDN [21]; and three diffusion model-based defense methods, GNS [50], DiffPure [51], and GDMP [52]. To compare our methods

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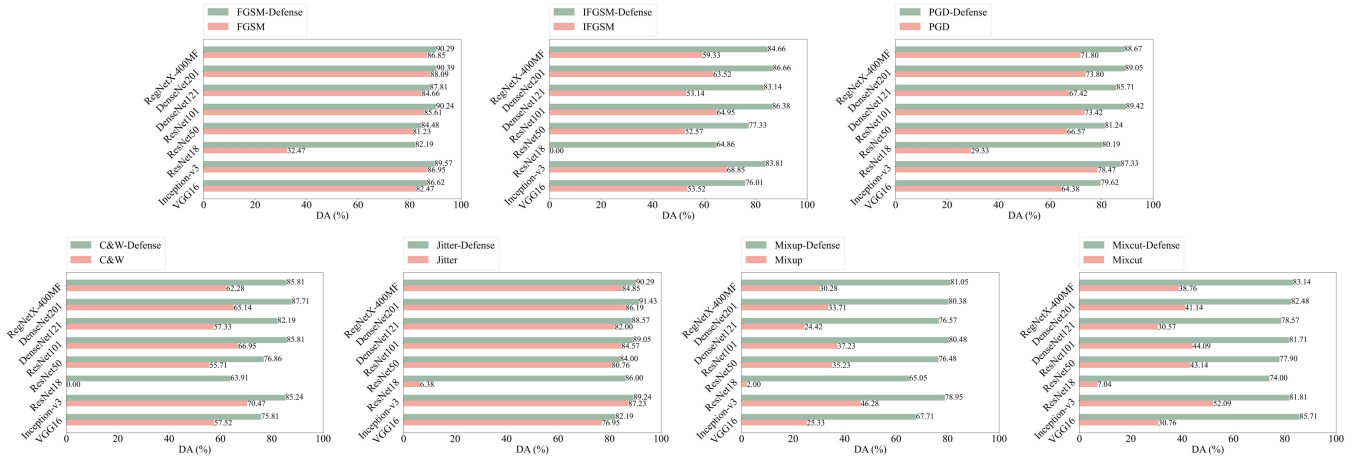


Fig. 3. Defense performance of our method under different attack methods in the UCM dataset.

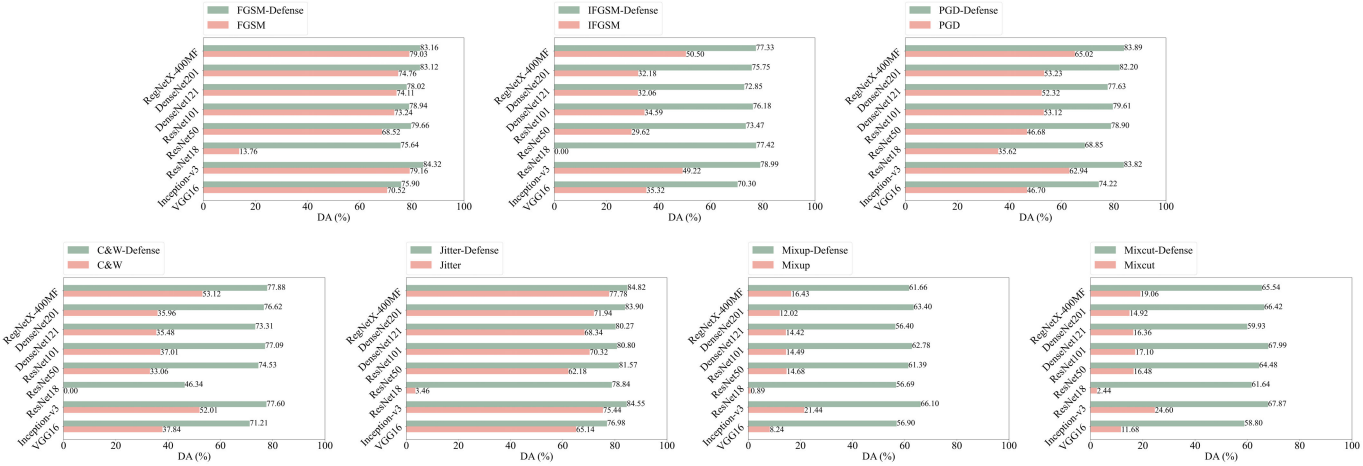


Fig. 4. Defense performance of our method under different attack methods in the AID dataset.

with these approaches, we established consistent experimental conditions.

The third part shows ablation experiments and hyperparametric sensitivity analysis. Finally, qualitative results are provided. Specifically, to investigate the changes in adversarial examples before and after the defense, we analyzed the pixel distributions of four types of samples and performed visualizations for better understanding.

C. Adversarial Defense

In this set of experiments, we evaluated the defensive capabilities of the proposed method against various attacks. The experimental results on the UCM dataset are shown in Fig. 3. It can be observed that seven attack methods led to a decrease in the target model's classification accuracy. FGSM, IFGSM, and PGD are perturbation-based attacks with relatively weak attack transferability. On the other hand, C&W and Jitter attack methods exhibited strong transferability, significantly reducing the classification accuracy of multiple target models. Mixcut and Mixup are feature-based attacks carefully designed by Xu and Ghamisi [32], rendering almost all target models ineffective. The entry labeled "*-Defense" indicates the defense performance after applying our defense method following the "*" attack. It can be seen that using our defense network significantly improves the classification accuracy of the target model, demonstrating the effectiveness of our approach against all attack methods and its ability to restore the target model's recognition capability.

From a data perspective, employing D3R-Net for defense not only enhances the classification accuracy on all adversarial examples but also elevates most of them to be above 75%, meeting the criteria for good usability. Otherwise, D3R-Net defense capabilities are treated equally for all models. Even when the target model is a source model such as ResNet18, D3R-Net protects it effectively. The AID and NWPU experimental results are presented in Figs. 4 and 5, respectively. Similar to the UCM dataset, the target model's classification accuracy is significantly improved after applying D3R-Net defense. After defense on the AID dataset, the classification accuracy of the target model is elevated chiefly to be above 60%. Compared to the UCM dataset, the AID dataset contains higher image resolutions, complex backgrounds, and more detailed information, making the recognition task for the target model more challenging.

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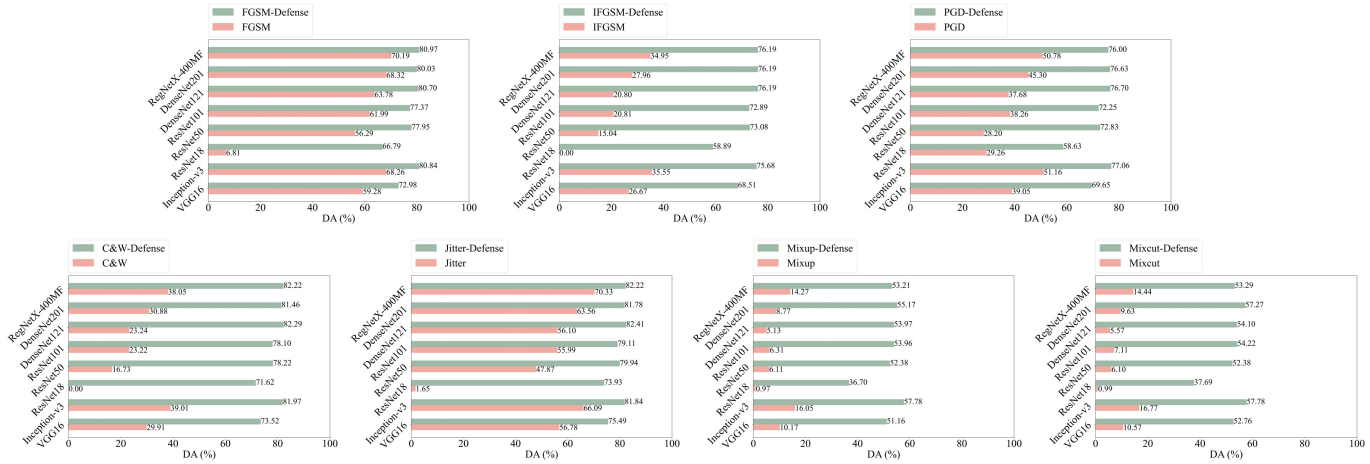


Fig. 5. Defense performance of our method under different attack methods in the NWPU dataset.

On the NWPU dataset, defense is effective against perturbation-based and optimization-based attack methods. However, for feature-based attack methods such as Mixup and Mixcut, the improvement in the target model's classification accuracy is relatively modest. It is attributed to the large number of categories, small inter-class distances, and weak separability of the NWPU dataset, resulting in limited defense effectiveness against specific attack methods.

D. Comparative Study

To evaluate the impact of different adversarial defense methods on adversarial example input, we tested various deep learning models' overall accuracy (OA) using different transformation techniques on the first publicly available remote sensing adversarial examples dataset, UAE-RS. Quantitative evaluations were conducted to assess the influence of these defense methods on classification performance, as shown in Tables II and III, where bold indicates the best and underlined indicates the second-best results. Due to the threat of adversarial attacks, existing deep learning models without transformation or defense methods struggle to achieve satisfactory recognition results.

1) Defense Performance Comparison: On the UAE-RS adversarial test set, all models used in this study achieved an OA of only 30% or below without any transformation. Furthermore, traditional filtering and transformation methods, such as Mean-Filter [46] and GB-Filter [47], yielded limited and unstable improvements. In some cases, their accuracy is even reduced as these methods may introduce new noise or style differences detrimental to model deployment.

APE-GAN [22] utilizes FGSM to generate adversarial samples from clean samples and then feeds the clean and adversarial samples into the discriminator and generator for training, respectively. This approach aims to achieve defense by reconstructing the adversarial samples using the generator. Experimental results show that the seed approach can mitigate a small portion of the adversarial noise but performs poorly against unknown attacks. Moreover, methods based on GAN architectures generally suffer from training instability. MAD-VAE [48] is an augmented iteration of Defense-VAE [53] that is trained using adversarial samples generated through a variety of attack strategies such as FGSM, IFGSM, and C&W. Experimental results show that MAD-VAE consistently outperforms the APE-GAN approach. This advantage can be attributed to the fact that MAD-VAE is exposed to various types of adversarial samples during training, allowing it to learn a broader range of adversarial noise distributions. In addition, MAD-VAE also exhibits higher training stability, which is a significant advancement in the field of GAN-based alternatives.

Deep learning-based denoising methods, such as HiNet [49] and TGDN [21], were demonstrated to have more significant performance improvements than traditional transformation methods. However, solely using traditional denoising networks offered limited performance enhancements and showed almost no defense effect on the ResNet18 model. Since traditional denoising methods aim to minimize the distance between adversarial examples and clean samples by attempting to remove surface noise from the images, they may not be effective in defending against adversarial attacks.

The methods based on the diffusion model are all mapping both adversarial and standard samples to Gaussian distribution through the forward noise addition process of the diffusion model, which is restored in the denoising process. The performance of the method is affected by the choice of parameters, and the choice of the time step τ is very critical; too small a time step may not be able to purify the noise effectively, and too long a time step may easily make the image lose the original distribution. The DiffPure [51] method sets the optimal time step to 150, which takes longer due to the execution of longer noise injection steps and possibly causes the image to lose the original distribution. GNS [50] is built on the DiffPure framework, which only removes the noise for a specified number of steps by optimizing the number of steps in the denoising process. This leads to difficulty effectively removing adversarial noise from the image and poor defense. GDMP [52] introduces conditional denoising, using SSIM or MSE to improve the similarity between the denoised image and the antagonistic image in the denoising process; thus, indirectly improving the similarity between

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TABLE II
DEFENSE ACCURACY (%) OF EIGHT DEEP MODELS ON THE UAE-RS UCM ADVERSARIAL TEST SET WITH DIFFERENT COMPARISON METHOD

Model	No Transform	Mean-Filter	GB-Filter	APE-GAN	MAD-VAE	HiNet	TGDN	GNS	DiffPure	GDMP	D3R-Net
VGG16	19.52	39.90	31.33	39.33	37.14	57.24	68.67	32.10	37.24	62.67	**77.48**
Inception-v3	30.19	56.00	47.42	58.19	66.10	64.76	74.48	43.14	46.00	63.62	**83.06**
ResNet18	2.95	16.85	7.80	10.57	14.86	4.10	9.90	9.43	25.33	41.24	**87.43**
ResNet50	25.53	44.76	42.28	27.14	33.62	52.38	65.81	25.81	42.57	53.14	**81.30**
ResNet101	28.10	49.52	36.38	44.67	52.76	56.48	69.43	50.67	66.48	70.86	**82.56**
DenseNet121	17.14	36.85	33.04	32.00	44.00	45.81	61.52	25.62	35.14	65.33	**83.37**
DenseNet201	26.38	52.09	38.57	37.90	41.43	51.33	64.95	41.14	57.14	68.76	**80.27**
RegNetX-400MF	27.33	43.62	36.57	56.00	55.05	56.67	66.38	44.57	49.81	63.52	**77.20**

TABLE III
DEFENSE ACCURACY (%) OF EIGHT DEEP MODELS ON THE UAE-RS AID ADVERSARIAL TEST SET WITH DIFFERENT COMPARISON METHOD

Model	No Transform	Mean-Filter	GB-Filter	APE-GAN	MAD-VAE	HiNet	TGDN	GNS	DiffPure	GDMP	D3R-Net
VGG16	11.88	23.59	14.60	19.60	26.00	47.78	51.34	17.92	18.50	41.92	**56.84**
Inception-v3	21.54	36.92	27.42	20.94	31.02	53.16	**65.14**	20.82	24.04	35.40	64.75
ResNet18	1.28	10.78	6.11	4.66	7.56	4.60	24.22	7.34	15.72	30.56	**62.16**
ResNet50	10.74	22.56	16.63	17.36	20.02	41.10	60.66	20.72	27.98	41.08	**65.38**
ResNet101	11.56	26.15	17.49	22.80	25.50	45.04	**66.90**	21.76	25.04	45.54	63.92
DenseNet121	10.30	20.51	15.12	19.84	22.56	39.28	58.16	16.66	26.42	43.36	**65.38**
DenseNet201	14.22	19.48	14.09	17.02	19.82	44.12	65.60	17.24	21.90	43.70	**67.32**
RegNetX-400MF	23.20	21.03	19.50	19.06	23.46	51.28	64.78	23.80	25.58	43.02	**66.15**

the denoised image and the original clean image, therefore alleviating the destruction of the original distribution of the denoising by the high time step τ. Thus, its antagonistic noise removal is better, and the defense effect is better. D3R-Net adopts an effective truncation strategy, and at the same time retains the semantic information of the denoised image during the denoising process using the DCFM module. D3R-Net uses an effective truncation strategy while maintaining the semantic information of the denoised image in the denoising process through the DCFM module to achieve optimal defense.

To provide a more intuitive assessment of the performance of various methods, we visualized some adversarial examples from the UAE-RS dataset. We presented the corresponding transformed images using different methods in Figs. 6 and 7. Compared to traditional filtering and transformation methods such as Mean-Filter and GB-Filter, the denoising methods generated transformed images that closely resemble the original clean samples in appearance. Generative defense methods APE-GAN and MAD-VAE have reconstructed adversarial samples that are still visually noisy and visible due to the limited amount of noise removed. Adversarial samples reconstructed by diffusion model-based methods are affected by the degree of denoising. Moreover, we observed that the denoised images generated by HiNet, TGDN, and D3R-Net are visually challenging for human observers. However, the denoising process, employing the DCFM, may lead to detail loss in the restored image compared to the ground truth (GT), as it filters not only adversarial noise but also image details. These detail losses have little effect on the model recognition performance and are visually imperceptible, which has been verified in the ablation experiment.

2) Time Cost and Computational Complexity Analysis: Table IV compares the results in terms of time cost, number of parameters, and computational complexity. The time cost was derived by averaging the total restoration time over 1050 images, while computational complexity was quantified based on a standard image size of $3 \times 256 \times 256$. Filter-based methods, characterized by direct processing that circumvents intricate deep model inference, boast the shortest processing intervals, typically under 0.01 s per frame, evidencing high efficiency. Generative models such as APE-GAN and MAD-VAE, with uncomplicated encoder–decoder frameworks, also register brief computation times. Meanwhile, HiNet and TGDN, adopting supervised

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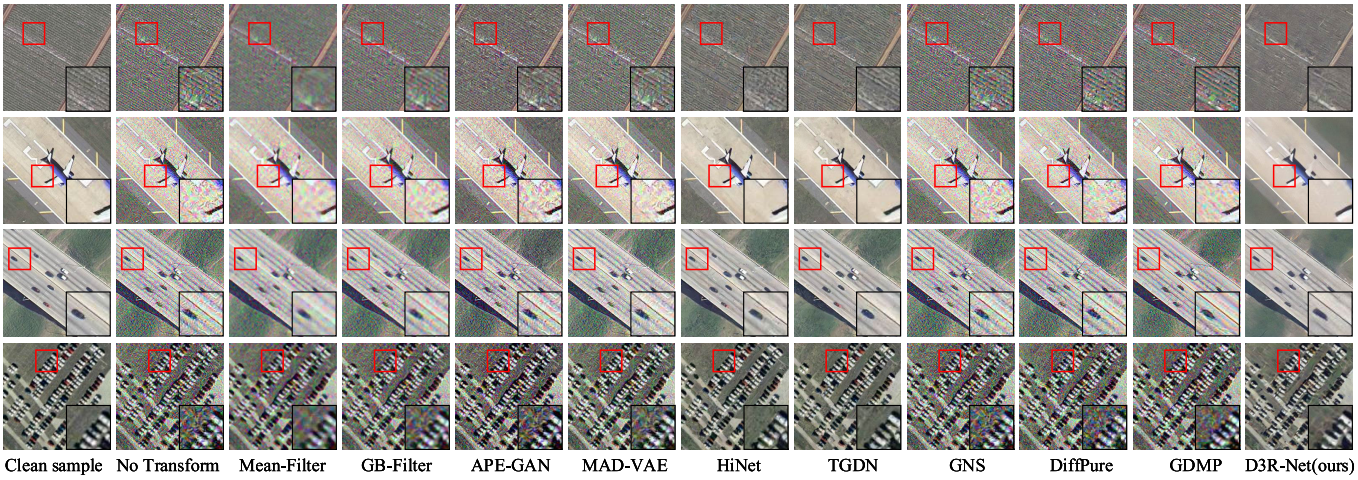


Fig. 6. Four random selected adversarial examples from the UAE-RS UCM dataset and corresponding images transformed using different methods.

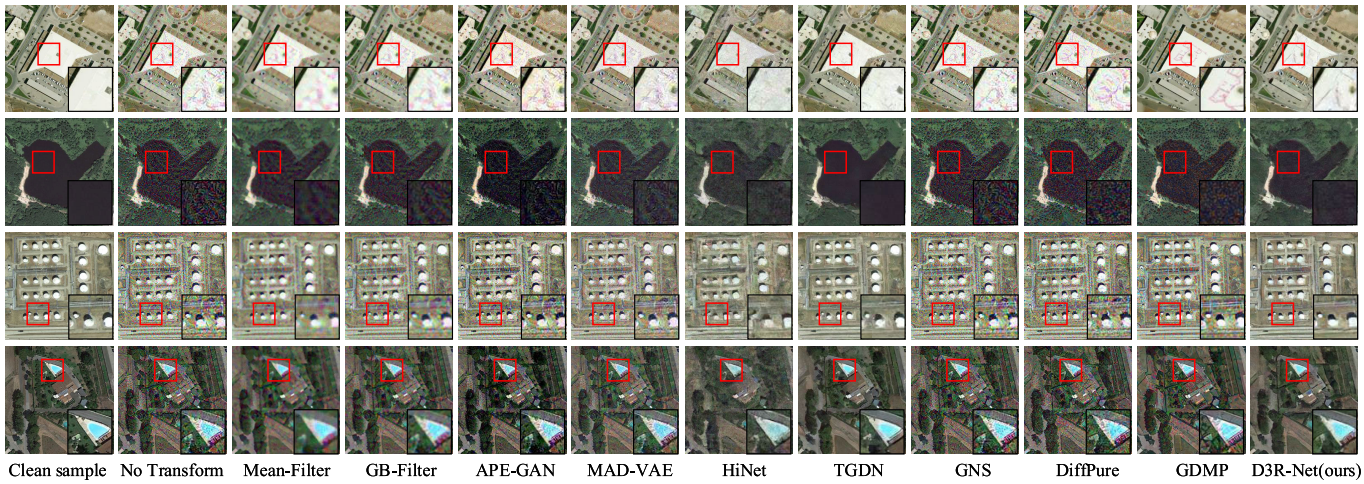


Fig. 7. Four random selected adversarial examples from the UAE-RS AID dataset and the corresponding images transformed using different methods.

TABLE IV
TIME COST AND PARAMETER AMOUNT COMPARISON ON THE UAE-RS

Method	Time Cost(S/F)	Params(MB)	FLOPs
Mean-filter	<0.01	—	—
GB-filter	<0.01	—	—
APE-GAN	0.03	17.26	40.20G
MAD-VAE	0.01	12.80	16.51G
HiNet	0.33	88.67	170.48G
TGDN	0.33	89.19	172.86G
GNS	8.21	552.66	1.11T
DiffPure	13.18	552.66	1.11T
GDMP	3.94	552.66	1.11T
D3R-Net	1.02	93.52	193.77G

learning methodologies and sharing an identical network architecture, exhibited highly similar processing times and resource consumption patterns. DiffPure and GNS are built on the same diffusion model framework while examining diffusion-model-based restoration techniques; DiffPure takes longer forward noise addition steps, leading to relatively prolonged time costs. GDMP uses the acceleration method in IDDPM to reduce the time step, resulting in a shorter time cost. D3R-Net also uses the U-Net structure, but with smaller model layers and parameter settings, in addition to effective control of the noise injection and denoising steps, thus reducing time costs and resource requirements.

Regarding the number of method parameters and computational complexity, APE-GAN and MAD-VAE are effective choices due to their smaller network structures, which significantly reduce resource consumption. Given their architectural homogeneity, HiNet and TGDN show nearly equivalent parameter counts and computational requirements. All diffusion-model-based approaches uniformly employ the standard U-Net model, resulting in high parameters and computational load consistency. Conversely, D3R-Net significantly reduces parameter counts and computational costs by implementing a more streamlined model configuration, further optimizing resource efficiency.

E. Ablation Study

1) Model Ablation Study: Table V shows the classification performance of adversarial examples generated using ResNet18 as the source model on VGG16, and ablation experiments were performed on three datasets. The first row shows

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TABLE V
PERFORMANCE CONTRIBUTION OF EACH MODULE OF THE D3R-NET TO THE SCENE CLASSIFICATION TASK

Attack	Diffusion	DCFM	UCM	AID	NWPU
			91.90	89.62	86.05
	✓		88.21	87.04	85.29
	✓	✓	89.36	88.01	85.47
✓			30.76	11.68	10.57
✓	✓		78.19 (↑47.43)	53.64 (↑41.96)	51.68 (↑41.11)
✓	✓	✓	85.71 (↑7.52)	58.80 (↑5.16)	52.76 (↑1.08)

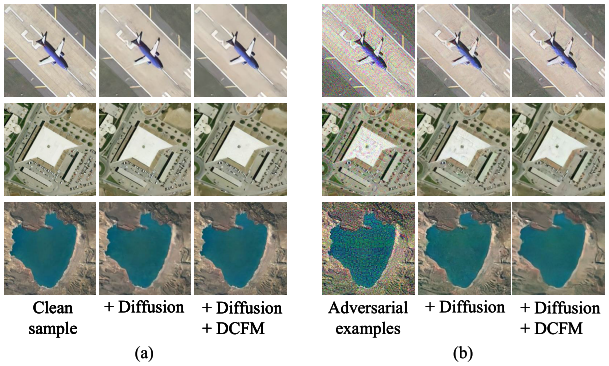


Fig. 8. Add each component to reconstruct the image. The three rows correspond to random samples selected on UCM, AID, and NWPU datasets. (a) Clean samples are defended, aiming to show the harmlessness of the D3R-Net. (b) Defense against adversarial examples.

TABLE VI
EFFECT OF τ VALUE ON DEFENSE PERFORMANCE

τ	14	15	16	17	18	19	20	21	22	23
UCM	79.33	80.67	82.38	**83.14**	82.29	81.44	80.38	78.95	77.62	76.57
AID	50.93	51.80	52.24	52.92	53.84	54.27	**56.29**	55.05	54.42	53.10
NWPU	55.30	56.63	57.97	58.16	58.67	59.62	60.06	**60.13**	60.00	58.67

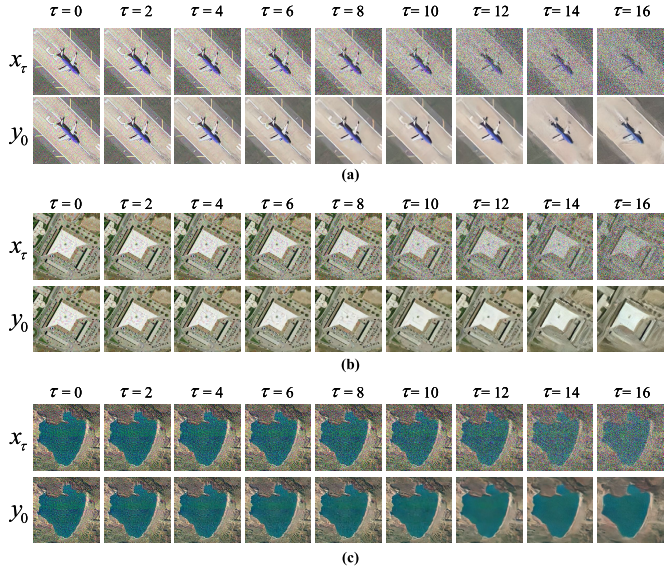


Fig. 9. Schematic of x_τ and reconstructed image y_0 at different depths τ. (a) Random sample of UCM. (b) Random sample of AID. (c) Random sample of NWPU.

the classification accuracy of clean samples on VGG16. The second and third rows show that after joining D3R-Net, which could reflect the side effects of D3R-Net on clean samples. The results show that D3R-Net can be added to classifier models with little negative impact on clean samples. Fig. 8(a) is a schematic of its reconstruction, showing the reconstructed image after adding different components of D3R-Net. The reconstructed image is almost indistinguishable from the clean sample. The last three lines in Table V show the accuracy after the Mixcut-Attack, the accuracy after adding the diffusion defense network, and the effect after adding the DCFM. Its reconstruction diagram is shown in Fig. 8(b). The results show that both D3R-Net and modules positively contribute to the overall defense performance and can effectively improve the classification performance of classifier models against adversarial examples.

2) Hyperparameter Sensitivity Study: In this study, the parameter τ represents the number of iterations, that is, the number of noises added. To analyze the impact of τ on performance, we experiment with different values of τ. When τ is 0, there is no defense, and the result is the classification accuracy of Mixcut-Attack. We performed a parameter search on the three datasets, specifically denoising starting from τ. In steps x_τ to y_0, we tested the output of each step to get the τ with the best result corresponding to it. It can be seen from Table VI that the DA on the three datasets first increases and then decreases with the increase of τ. This is because when τ is small, the number of diffusion steps is small. The distribution difference between adversarial examples and original samples is still huge, which is not conducive to restoration. At the same time, as τ increases, the noise image gradually becomes a pure noise image, resulting in more loss of semantic information, which is also not conducive to image restoration. Fig. 9 shows the corresponding x_τ for different values of τ on the three datasets and the reconstructed image y_0 when denoising from step τ to step 0. In addition, the three datasets have three optimal τ values with little difference. The main reason is that the three datasets have specific differences. For example, compared with the UCM dataset, the AID dataset has higher pixel resolution and more complex background information. In contrast, the NWPU dataset has the most significant number of categories and remarkable similarity between categories.

F. Qualitative Results and Analysis

To better demonstrate the data reconstruction effect of our method, as shown in Fig. 10, we randomly select three categories from the UCM dataset for visualization, namely airplane, freeway, and parkinglot, and each row corresponds to a category. We analyzed the pixel distributions of samples from these categories. In the visual presentation, the first column

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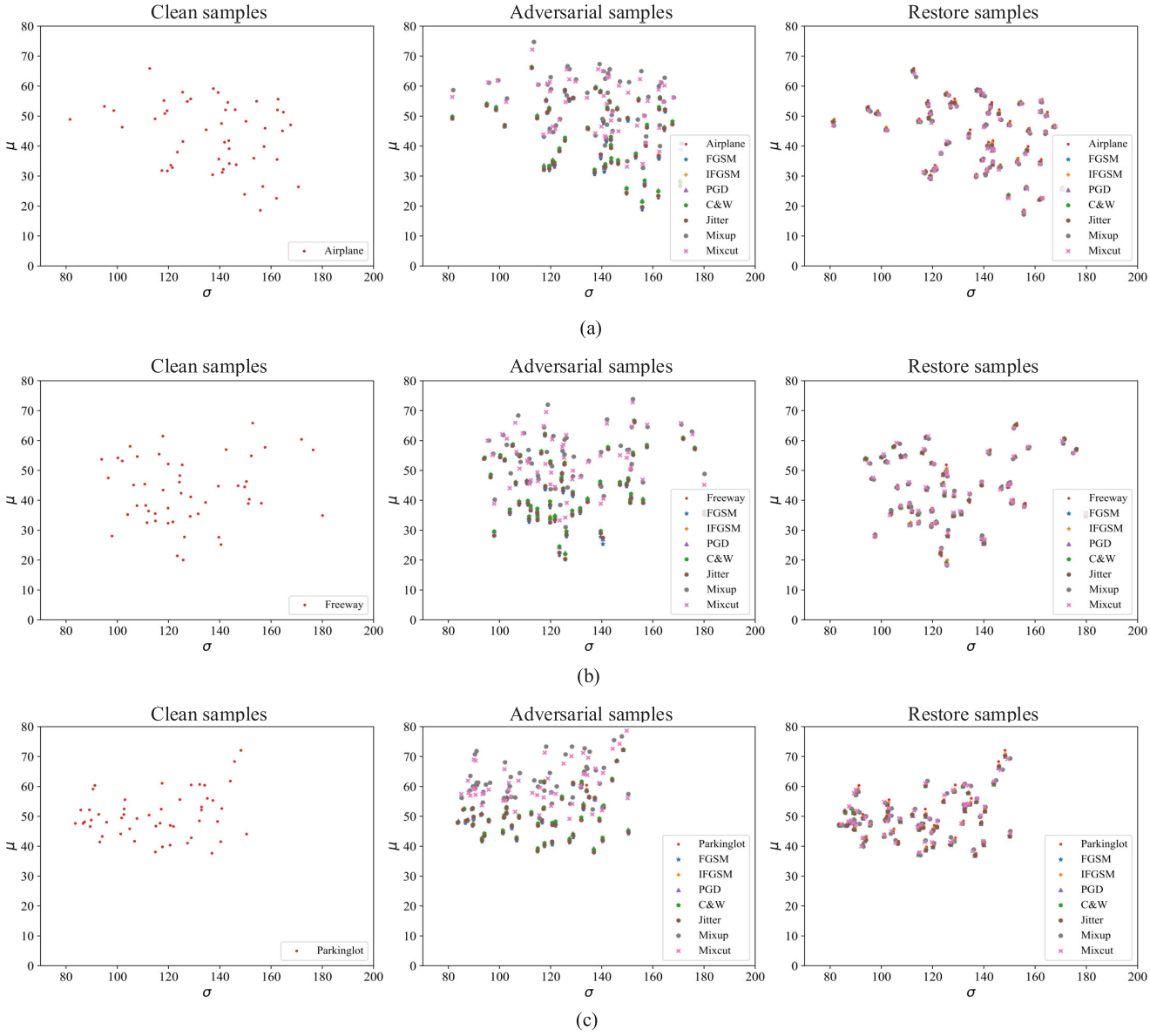


Fig. 10. Qualitative results on three categories, the first column shows the distribution of clean samples in three randomly selected categories, the second column shows the pixel distribution of adversarial samples generated using seven adversarial attack methods, and the third column shows the pixel distribution of restore samples after using D3R-Net. (a) Category—Airplane. (b) Category—Freeway. (c) Category—Parkinglot.

displays the pixel distribution of clean samples for these three categories, and the middle column shows the pixel distribution of adversarial examples after undergoing seven different attack methods. The pixel distribution of the adversarial samples generated by the seven attack methods is shifted compared with the clean samples, which is due to small perturbations. In addition, the stronger the attack capability of the adversarial sample, the greater the degree of pixel shift, such as Mixcut.

The last column illustrates the pixel distribution of the restored samples after passing through D3R-Net. From the results, D3R-Net successfully restores the pixel distribution of adversarial examples to a state consistent with the clean samples, enabling the target model to perform recognition tasks correctly. Such data reconstruction is crucial in enhancing the model's robustness and defending against adversarial attacks. Through our defense method, the interference caused by adversarial examples is effectively eliminated, and the original structure and features of the data are restored, allowing the target model to perform more accurate classification tasks. These visualization results further support the effectiveness and reliability of D3R-Net in adversarial environments. Furthermore, D3R-Net is adaptable to deal with seven typical attackers effectively and successfully recovers samples corrupted by various adversarial strategies, demonstrating its robustness in defending against various adversarial attack strategies.

V. Conclusion

Despite achieving advanced performance in interpreting Earth sciences and remote sensing data, deep learning methods exhibit vulnerability to adversarial examples in practical applications. This article proposes an effective D3R-Net to resist the threats posed by adversarial examples in remote sensing scene classification tasks. Specifically, it aims to restore adversarial examples to clean ones by leveraging their initial

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semantic consistency with clean samples and their consistency in distribution after diffusion. The UAE-RS dataset is used in the experiments to further evaluate the impact of D3R-Net on the classification results of different deep learning models. Extensive experimentation demonstrates that D3R-Net significantly enhances the resistance of various deep learning models against adversarial examples.

Compared with existing defense methods, D3R-Net does not need to retrain the model or eliminate suspicious samples. In addition, it can be used offline or online without harming the original model, so it is more versatile and practical. Compared with depth-based image restoration methods, D3R-Net focuses on the restoration of global discriminative features, can better defend against unknown attacks, and more importantly, does not require labels during training.

The D3R-Net primarily focuses on recovering global discriminative information and exhibits limitations in restoring fine-grained details. In the next work, we will integrate the proposed network with convolutional neural networks to capitalize on its robust learning capability in restoring image details. We anticipate more robust defense performance by supplementing these discriminative details.

In future work, there will be a broad space for exploration in remote sensing scene classification and adversarial defense research. It is worth noting that, given the inherent spatial characteristics of remote sensing scene classification and the need for multisource data fusion, future research should integrate spatial information theory and multimodal data processing technology to develop a set of adversarial attack defense solutions based on spatial context. This solution may enhance the robustness of the model in dealing with different spatial scales and multiscale scenarios, ensuring that the remote sensing scene classification system shows higher stability and security in actual complex environments. In addition, the application expansion of cross-domain counterattack and defense technologies is also a promising research direction.

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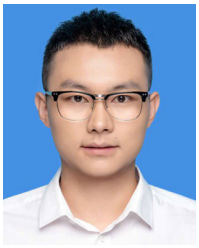
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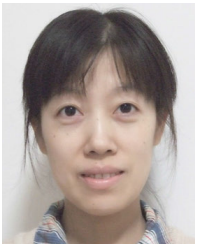
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